自动驾驶闭环测试的现状

Current Status of Closed-Loop Testing for Autonomous Driving

中文版本

1 问题背景

自动驾驶系统的核心目标不是在离线数据集上预测一条看起来接近人类驾驶员的轨迹，而是在真实交通环境中持续、稳定、安全地完成驾驶任务。因此，自动驾驶评价的关键不应只停留在 open-loop prediction 或 imitation error，而应进入 closed-loop testing：系统的动作会改变自车状态，自车状态会改变下一时刻观测，新的观测又会影响后续决策。

然而，当前自动驾驶研究存在一个明显断层：真实道路数据越来越多，但这些数据大多以离线 log 的形式存在。它们非常适合做感知、检测、跟踪、预测或者 open-loop planning evaluation，却很难直接用于驾驶闭环测试。换句话说，真实数据记录的是历史上真实 ego vehicle 走过的轨迹；一旦被测试模型输出了不同的轨迹，原始 log 中对应的新相机图像、LiDAR 点云和周围交通参与者反应就不存在了。

因此，当前自动驾驶闭环测试的核心问题可以概括为：

> 如何将真实车辆运行数据转化为可交互、可闭环、可评价的测试环境，使端到端自动驾驶系统能够在接近真实世界分布的传感器输入下进行长期闭环测试？

这个问题的价值在于：它连接了真实数据规模化利用、端到端自动驾驶评价、长尾场景构造以及安全验证。它不是简单地做一个更漂亮的仿真器，也不是再定义一个离线指标，而是要解决“真实数据如何变成闭环世界”的问题。

2 当前闭环测试的主要类型

当前自动驾驶测试大致可以分为四类。

2.1 开环真实数据测试

典型做法是使用真实传感器输入，让模型输出未来轨迹，然后与人类专家轨迹进行比较，例如计算 ADE、FDE 或 planning L2 error。这类方法的优点是输入真实、评估成本低、数据规模可以很大；缺点是模型的输出不会影响未来输入，因此不能测试误差累积、轨迹纠偏、长时控制稳定性和交互能力。

开环测试本质上是 teacher-forcing evaluation。模型每一帧看到的仍然是真实 log 中的状态，而不是自己上一时刻动作导致的新状态。因此，开环表现好并不代表闭环驾驶能力强。

2.2 规划层闭环测试

nuPlan 和 Waymax 代表了这一类方法。它们基于真实道路数据或真实 motion data，构建 object-level 或 map-level 的闭环测试环境。被测 planner 接收 ego state、HD map、周围 object states 或 tracks，然后输出轨迹；仿真器推进 ego 状态，并用碰撞、进度、舒适性、TTC 等指标评价规划性能。

这类方法的核心假设是：

感知结果已经以较高质量的 object states、tracks 和 semantic map 形式给出。

因此，它们更适合评价 planner，而不是端到端系统。其优点是闭环机制相对清晰、计算效率高、评价指标更接近驾驶行为；缺点是它默认感知已经基本解决，无法评价 camera/LiDAR 到驾驶动作之间的信息损失、感知误差传播和 sensor distribution shift。

2.3 非反应式端到端伪闭环测试

NAVSIM 是这一类方法的代表。它引入真实 sensor input，让端到端模型基于 camera、LiDAR 或 ego status 预测未来 ego trajectory。随后，它并不重新生成下一帧传感器输入，而是在 BEV abstraction 中短时展开这条轨迹，并根据 collision、drivable area、TTC、progress、comfort 等指标计算 PDM Score 或其变体。

这种方法比传统开环 L2 指标更进一步，因为它不再只问“轨迹像不像人类”，而是问“这条轨迹是否可驾驶”。但是，它仍然不是严格意义上的闭环仿真。原因是环境不根据 ego 行为响应，模型也不会在执行一步后接收新的 sensor observation 并继续决策。

因此，NAVSIM 可以被理解为：

真实传感器输入 + 一次性轨迹预测 + 非反应式 BEV 几何评价。

它的优点是规模大、成本低、适合 leaderboard；缺点是可能给人一种“已经接近闭环评价”的错觉。对于真正的端到端闭环测试而言，它绕开了最关键的问题：ego 偏离 log 后，新的世界状态和传感器观测如何生成。

2.4 传感器级闭环仿真

VISTA 2.0、UniSim、HUGSIM、DriveArena 和 DrivingSphere 代表了更接近端到端闭环测试的方向。它们试图解决的问题是：当 ego vehicle 偏离原始 log 后，如何生成新的 camera、LiDAR 或 multi-view video observation，从而让端到端模型可以持续 observe–act–observe。

其中，VISTA 2.0 是较早的真实数据驱动多模态传感器仿真器，支持 RGB camera、LiDAR、event camera 等输入，并可用于 perception-to-control policy learning。UniSim 进一步提出将单条实车 recorded log 转化为 realistic closed-loop multi-sensor simulation，通过 neural feature grids 重建静态背景和动态 actor，并在新视角、新 actor placement 下合成 camera 和 LiDAR 数据。HUGSIM、DriveArena 和 DrivingSphere 则进一步引入 3D Gaussian Splatting、conditional generative model、4D occupancy world 和 multi-view video synthesis 等技术，试图提升视觉真实感和闭环可交互性。

这一方向最接近本文关注的问题，但仍未完全解决。当前主要瓶颈包括：动态交通参与者的真实交互建模不足，ego 大幅偏离 log 后的未观测区域补全困难，多模态传感器物理不完整，长时闭环稳定性难验证，生成式输入可能引入新的 domain gap。

3 两条主要技术路线

围绕端到端闭环测试，可以抽象出两条路线。

3.1 路线一：从仿真器出发，使仿真输入逼近真实

这一路线以 CARLA 等 synthetic simulator 为代表。其基本思路是：

> Synthetic world → realistic rendering → realistic sensor simulation → closed-loop testing.

这条路线的优点是闭环天然成立。仿真器中有完整的世界状态，自车动作可以推进物理世界，下一帧传感器观测可以直接渲染出来。它还具有高可控性，可以方便地设置天气、交通流、道路结构和危险场景，并且可以获得完整 ground truth。

但是，它的核心问题是 domain gap。真实世界中的相机成像、LiDAR 反射、radar 噪声、光照、材质、路面磨损、动态交通行为、人类驾驶习惯、长尾场景分布都很难被 synthetic simulator 完整逼近。对于端到端系统而言，输入分布的偏差可能直接影响感知表征和后续规划行为。

因此，这条路线的问题不是不能闭环，而是闭环的世界可能不够真实。

3.2 路线二：从真实数据出发，将真实数据转化为闭环仿真器

这一路线以 VISTA 2.0 和 UniSim 为核心代表。其基本思路是：

> Real-world driving logs → scene reconstruction → editable world representation → regenerated sensor observation → closed-loop testing.

这条路线的优势在于输入起点是真实的。真实道路布局、真实传感器特性、真实光照、真实交通分布和真实长尾样本都来自实车数据。相比于从 synthetic world 出发逼近真实，它不需要从零构造世界的视觉外观，而是需要解决真实 log 在 ego 偏离后的世界演化问题。

这也是本文更建议推进的路线。因为当前车企和数据集已经积累了大量真实道路数据，真正的瓶颈不是没有真实输入，而是这些真实输入缺乏可交互闭环能力。将真实数据转化为闭环仿真器，可以更直接地服务端到端自动驾驶评价。

当然，这条路线也有难点：真实 log 只记录了一个已经发生的世界；一旦 ego 或其他 actor 被修改，就需要补全新的遮挡关系、动态响应、未观测区域、传感器噪声和时序一致性。这本质上需要一个 grounded world model，而不是简单的数据重放。

4 方法对比

5 端到端自动驾驶技术暴露出的本质问题

当前端到端自动驾驶技术虽然在结构设计上已经从简单的 image-to-steering 发展到更复杂的 structured end-to-end framework，例如感知、跟踪、建图、预测、occupancy 和 planning 的联合建模，但其训练和评价仍然高度依赖 open-loop 数据。

这带来一个根本问题：

类型	代表方法	输入是否真实	闭环程度	核心不足
传统开环测试	nuScenes-style, open-loop planning	真实	无闭环	只评价预测是否接近专家轨迹，不能测试误差累积和恢复能力。
规划层闭环	nuPlan, Waymax	object/map 真实或离线标注	planner-level closed-loop	默认感知结果已知，无法评价 sensor-to-action 的端到端闭环能力。
伪闭环端到端测试	NAVSIM	真实 sensor input	non-reactive pseudo-simulation	不生成下一帧 sensor observation，环境不响应 ego，可能高估闭环能力。
Synthetic closed-loop	CARLA	仿真输入	full closed-loop	世界可控但 domain gap 大，传感器物理和真实交通分布难以逼近。
真实数据传感器级闭环	VISTA 2.0, UniSim	真实 log 重建	sensor-level closed-loop	可编辑范围、未观测区域补全、动态交互和长时稳定性仍有限。
生成式视觉闭环	HUGSIM, DriveArena, DrivingSphere	部分真实数据或生成式重建	visual closed-loop	视觉生成进步明显，但交互动力学、多模态传感器和评估可信度仍不足。

表 1: 自动驾驶闭环测试路线对比

> 端到端模型往往是在 expert trajectory 附近学会了预测合理轨迹，但并没有被充分训练和验证为一个能够长期稳定闭环运行的驾驶策略。

因此，开环性能和闭环性能在原理上存在巨大差异。开环评估中，模型在当前时刻的错误不会影响下一帧输入；闭环评估中，一个很小的轨迹偏差会改变 ego pose，进而改变下一帧 observation，最终导致 distribution shift、误差累积、失控或碰撞。

这也解释了为什么一些端到端模型在开环 benchmark 上表现较好，但一进入闭环仿真，route completion、collision rate、off-road rate 等指标可能显著恶化。问题不只是模型本身，也包括评价环境是否真实、是否可交互、是否能可信地生成下一时刻 sensor observation。

6 本文建议的研究方向

基于以上分析，更有价值的研究方向不是继续构造一个更大的 open-loop benchmark，也不是仅仅优化 synthetic simulator 的视觉真实感，而是：

> 从真实车辆运行数据出发，构建端到端可用的传感器级闭环仿真测试平台。

这一路线的关键思想是：既然真实传感器输入已经存在，那么不必从零构造一个完全 synthetic 的世界；更合理的做法是把真实数据转化为可交互的世界模型。这样，研究重点可以集中在最核心的问题上：

1. ego 偏离原始 log 后，静态背景如何从新视角渲染；
2. 动态 actor 如何根据 ego 行为产生合理响应；
3. 遮挡和未观测区域如何补全；
4. camera、LiDAR、radar 等传感器输出如何保持物理一致性；
5. 长时闭环下如何评价误差累积、恢复能力和安全风险；
6. 生成式世界是否与真实道路风险分布保持统计关联。

因此，本文建议的核心研究命题可以表述为：

> Real-world-data-based sensor-level closed-loop simulation for end-to-end autonomous driving evaluation.

中文可以表述为：

> 基于真实车辆运行数据的端到端自动驾驶传感器级闭环测试环境构建。

7 结论

自动驾驶闭环测试的现状可以概括为：CARLA 等 synthetic simulator 有闭环但不够真实；nuPlan 和 Waymax 基于真实数据但主要评估 planner，并默认感知先验可用；NAVSIM 引入真实传感器输入但仍是 non-reactive pseudo-simulation；VISTA 2.0 和 UniSim 开始真正走向真实数据驱动的传感器级闭环仿真；HUGSIM、DriveArena 和 DrivingSphere 则代表了生成式视觉闭环仿真的新趋势。

当前最关键的空白是：如何将大规模真实车队数据转化为能够支持端到端模型长期闭环测试的交互式世界。这个问题直接决定了端到端自动驾驶技术能否从开环指标走向真实闭环能力验证。

因此，相比于“让仿真器更像真实”，更值得推进的方向是“让真实数据变成仿真器”。前者需要同时构建真实外观、真实传感器和真实交通动力学；后者已经拥有真实输入基础，核心难点集中在世界变化、交互响应和新观测生成上。这个方向更贴近车企真实数据利用，也更贴近端到端自动驾驶评价的本质需求。

English Version

8 Background

The ultimate goal of autonomous driving is not to predict a trajectory that is close to a human driver's trajectory in an offline dataset. The real goal is to drive continuously, safely, and robustly in interactive traffic environments. Therefore, the evaluation of autonomous driving systems should not stop at open-loop prediction or imitation error. It should move toward closed-loop testing, where the system's action changes the ego state, the changed ego state changes the next observation, and the new observation further affects future decisions.

However, there is a clear gap in current autonomous driving research. Large-scale real-world driving data are available, but most of them are stored as offline logs. Such logs are highly useful for perception, detection, tracking, prediction, and open-loop planning evaluation, but they are difficult to use for driving-level closed-loop testing. Once the evaluated model produces a trajectory different from the logged ego trajectory, the corresponding future camera images, LiDAR point clouds, and reactions of surrounding agents no longer exist in the original log.

Thus, the central problem can be formulated as follows:

> How can real-world driving logs be transformed into interactive, closed-loop, and evaluable simulation environments, so that end-to-end autonomous driving systems can be tested under realistic sensor inputs and long-horizon feedback?

The value of this problem lies in its connection to large-scale real-world data utilization, end-to-end driving evaluation, long-tail scenario construction, and safety validation. It is not merely about building a visually better simulator or defining another offline metric. The key issue is how to turn real-world data into a closed-loop world.

9 Main Categories of Current Closed-Loop Testing

9.1 Open-loop Real-data Evaluation

The common practice is to feed real sensor inputs to a model, let the model predict a future trajectory, and compare it with the human expert trajectory using metrics such as ADE, FDE, or planning L2 error. This setting has real inputs, low evaluation cost, and good scalability. However, the model output does not affect future inputs. Therefore, it cannot evaluate error accumulation, recovery behavior, long-horizon control stability, or interactive decision-making.

Open-loop evaluation is essentially teacher-forcing evaluation. At each frame, the model still observes the logged real-world state rather than the state caused by its previous action. As a result, strong open-loop performance does not necessarily indicate strong closed-loop driving capability.

9.2 Planner-level Closed-loop Testing

nuPlan and Waymax are representative examples of planner-level closed-loop testing. They use real-world driving data or real motion data to construct object-level or map-level simulation

environments. The planner receives ego states, HD maps, and surrounding object states or tracks, outputs a trajectory, and the simulator rolls out the ego state. Driving performance is evaluated using metrics such as collision, progress, comfort, time-to-collision, and route completion.

The key assumption is:

> Perception has already been converted into high-quality object states, tracks, and semantic maps.

Therefore, these benchmarks are more suitable for evaluating planners than end-to-end systems. Their advantage is that the closed-loop mechanism is clean and efficient, and their metrics are closer to driving behavior. Their limitation is that they assume perception is already solved, and thus cannot evaluate information loss, perception error propagation, or sensor distribution shift from camera/LiDAR to final driving actions.

9.3 Non-reactive End-to-end Pseudo-simulation

NAVSIM is representative of this category. It introduces real sensor inputs and asks end-to-end models to predict an ego trajectory from camera, LiDAR, or ego status. Then, instead of regenerating future sensor observations, it unrolls the predicted trajectory in a BEV abstraction and computes metrics such as collision, drivable area compliance, time-to-collision, progress, and comfort.

This is an improvement over traditional open-loop L2 evaluation because it asks not only whether the trajectory is close to the human trajectory, but also whether the trajectory is drivable. However, it is still not true closed-loop simulation. The environment does not react to ego behavior, and the model does not receive newly generated sensor observations after executing an action.

Therefore, NAVSIM can be summarized as:

> Real sensor input + one-shot trajectory prediction + non-reactive BEV geometric evaluation.

Its advantage is scalability, low cost, and suitability for leaderboard-style evaluation. Its limitation is that it may create the impression of being close to closed-loop testing while bypassing the most critical question: after the ego deviates from the log, how should the new world state and sensor observations be generated?

9.4 Sensor-level Closed-loop Simulation

VISTA 2.0, UniSim, HUGSIM, DriveArena, and DrivingSphere are closer to end-to-end closed-loop testing. They aim to answer the following question: when the ego vehicle deviates from the original log, how can we generate new camera, LiDAR, or multi-view video observations so that an end-to-end model can continuously operate in an observe–act–observe loop?

VISTA 2.0 is an early real-data-driven multi-sensor simulator supporting RGB cameras, LiDAR, and event cameras for perception-to-control policy learning. UniSim further converts a

single recorded log captured by a sensor-equipped vehicle into a realistic closed-loop multi-sensor simulation. It uses neural feature grids to reconstruct static backgrounds and dynamic actors, and synthesizes camera and LiDAR data from new viewpoints and new actor placements. HUGSIM, DriveArena, and DrivingSphere further introduce 3D Gaussian Splatting, conditional generative models, 4D occupancy worlds, and multi-view video synthesis to improve visual realism and closed-loop interactivity.

This direction is closest to the problem of interest. Nevertheless, it is still far from being fully solved. Major bottlenecks include realistic interaction modeling of dynamic agents, completion of unobserved regions when the ego deviates significantly from the log, incomplete multi-modal sensor physics, lack of long-horizon closed-loop validation, and new domain gaps introduced by generative observations.

10 Two Main Technical Routes

10.1 Route 1: Making Synthetic Simulation Inputs More Realistic

This route is represented by simulators such as CARLA. The basic idea is:

> Synthetic world → realistic rendering → realistic sensor simulation → closed-loop testing.

The advantage is that closed-loop simulation is naturally supported. The simulator maintains a complete world state, ego actions can update the physical world, and future sensor observations can be rendered accordingly. It also provides controllability over weather, traffic, road structures, and safety-critical scenarios, with privileged ground truth.

However, the core issue is the domain gap. It is difficult to fully match real-world camera imaging, LiDAR reflection, radar noise, illumination, material properties, road wear, traffic behavior, human driving habits, and long-tail scenario distributions. For end-to-end systems, such input distribution gaps can directly affect learned representations and downstream driving behavior.

Thus, the problem of this route is not the lack of closed-loop capability. Rather, the closed-loop world may not be sufficiently real.

10.2 Route 2: Turning Real-world Data into Closed-loop Simulators

This route is represented by VISTA 2.0 and UniSim. The basic idea is:

> Real-world driving logs → scene reconstruction → editable world representation → regenerated sensor observation → closed-loop testing.

The key advantage is that the starting point is real. Real road layout, real sensor characteristics, real illumination, real traffic distribution, and real long-tail samples are already present in the data. Compared with starting from a synthetic world, this route does not need to construct the visual appearance of the world from scratch. Instead, it needs to solve how the real logged world evolves after the ego deviates from the original trajectory.

This is the route recommended here. Since companies and datasets already contain massive real-world driving data, the central bottleneck is not the lack of realistic inputs, but the lack of interactive closed-loop capability. Turning real data into closed-loop simulators directly serves the evaluation of end-to-end autonomous driving systems.

Of course, this route is also challenging. A real log records only one world history. Once the ego or other actors are modified, the simulator must complete new occlusions, dynamic responses, unseen regions, sensor noise, and temporal consistency. This requires a grounded world model rather than simple log replay.

11 Comparison of Methods

Category	Examples	Input Realism	Closed-loop Level	Key Limitation
Open-loop testing	nuScenes-style, open-loop planning	Real	None	Evaluates similarity to expert trajectories; cannot test error accumulation or recovery.
Planner-level closed-loop	nuPlan, Waymax	Real or offline-labeled object/map states	Planner-level closed-loop	Assumes perception is already given; cannot evaluate sensor-to-action closed-loop behavior.
Pseudo-simulation	NAVSIM	Real sensor input	Non-reactive pseudo-simulation	Does not generate future sensor observations; the environment does not react to ego actions.
Synthetic closed-loop	CARLA	Synthetic input	Full closed-loop	Controllable but suffers from domain gap in appearance, sensor physics, and traffic distribution.
Real-data sensor-level closed-loop	VISTA 2.0, UniSim	Reconstructed from real logs	Sensor-level closed-loop	Limited editability, unseen-region completion, dynamic interaction, and long-horizon stability.
Generative visual closed-loop	HUGSIM, DriveArena, DrivingSphere	Partly real-data-based or generative	Visual closed-loop	Improved visual realism, but interaction dynamics, multi-modal sensors, and evaluation trustworthiness remain insufficient.

表 2: Comparison of autonomous driving closed-loop testing paradigms

12 What This Reveals about End-to-end Autonomous Driving

Modern end-to-end autonomous driving methods have evolved from simple image-to-steering models to structured end-to-end frameworks that jointly model perception, tracking, mapping, prediction, occupancy, and planning. However, their training and evaluation are still heavily based on open-loop data.

This creates a fundamental issue:

> End-to-end models may learn to predict reasonable trajectories near expert trajectories, but they are not sufficiently trained or validated as stable long-horizon closed-loop driving policies.

Therefore, open-loop and closed-loop performances are fundamentally different. In open-loop evaluation, an error at the current step does not affect the next input. In closed-loop evaluation, even a small trajectory error changes the ego pose, then changes the next observation, and may eventually lead to distribution shift, error accumulation, loss of control, or collision.

This explains why some end-to-end models perform well on open-loop benchmarks but degrade significantly in closed-loop simulation in terms of route completion, collision rate, and off-road rate. The problem is not only the model itself, but also whether the evaluation environment is realistic, interactive, and capable of generating trustworthy next-step sensor observations.

13 Recommended Research Direction

Based on the above analysis, the most valuable direction is not to build another large-scale open-loop benchmark, nor merely to improve the visual realism of synthetic simulators. The more important direction is:

> Constructing sensor-level closed-loop testing platforms from real-world vehicle operation data.

The key idea is that real sensor inputs already exist. Therefore, instead of building a fully synthetic world from scratch, it is more reasonable to transform real-world data into an interactive world model. The research focus can then be concentrated on the core missing components:

1. how to render static backgrounds from new ego viewpoints after deviation from the log;
2. how dynamic actors should respond to ego behavior;
3. how to complete occlusions and unseen regions;
4. how to maintain physical consistency for camera, LiDAR, radar, and other sensor outputs;
5. how to evaluate error accumulation, recovery ability, and safety risks in long-horizon closed-loop testing;

6. how to preserve statistical connections between generated scenarios and real-world risk distributions.

The research topic can be stated as:

> Real-world-data-based sensor-level closed-loop simulation for end-to-end autonomous driving evaluation.

14 Conclusion

The current status of autonomous driving closed-loop testing can be summarized as follows. CARLA-like synthetic simulators support full closed-loop testing but suffer from domain gap. nuPlan and Waymax use real data but mainly evaluate planners under the assumption that perception is already available. NAVSIM introduces real sensor inputs but remains a non-reactive pseudo-simulation. VISTA 2.0 and UniSim represent the beginning of real-data-driven sensor-level closed-loop simulation. HUGSIM, DriveArena, and DrivingSphere represent the emerging trend of generative visual closed-loop simulation.

The most important gap is how to transform large-scale real-world fleet data into interactive worlds that support long-horizon closed-loop testing of end-to-end driving models. This gap determines whether end-to-end autonomous driving can move beyond open-loop metrics toward real closed-loop capability validation.

Therefore, compared with "making simulators more realistic," a more promising direction is "turning real data into simulators." The former must construct realistic appearance, realistic sensors, and realistic traffic dynamics simultaneously. The latter already starts from real inputs and mainly needs to solve world evolution, interaction response, and new observation generation. This direction is better aligned with real-world fleet data utilization and the essential evaluation needs of end-to-end autonomous driving.

参考文献

[1] A. Dosovitskiy, G. Ros, F. Codevilla, A. Lopez, and V. Koltun, "CARLA: An Open Urban Driving Simulator," Conference on Robot Learning, 2017.

[2] H. Caesar et al., "nuPlan: A Closed-loop ML-based Planning Benchmark for Autonomous Vehicles," arXiv:2106.11810, 2021.

[3] C. Gulino et al., "Waymax: An Accelerated, Data-Driven Simulator for Large-Scale Autonomous Driving Research," NeurIPS Datasets and Benchmarks, 2023.

[4] D. Dauner et al., "NAVSIM: Data-Driven Non-Reactive Autonomous Vehicle Simulation and Benchmarking," arXiv:2406.15349, 2024.

[5] A. Amini et al., “VISTA 2.0: An Open, Data-driven Simulator for Multimodal Sensing and Policy Learning for Autonomous Vehicles,” IEEE International Conference on Robotics and Automation, 2022.

[6] Z. Yang et al., “UniSim: A Neural Closed-Loop Sensor Simulator,” IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023.

[7] H. Zhou et al., “HUGSIM: A Real-Time, Photo-Realistic and Closed-Loop Simulator for Autonomous Driving,” arXiv:2412.01718, 2024.

[8] X. Yang et al., “DriveArena: A Closed-loop Generative Simulation Platform for Autonomous Driving,” arXiv:2408.00415, 2024.

[9] T. Yan et al., “DrivingSphere: Building a High-fidelity 4D World for Closed-loop Simulation,” arXiv:2411.11252, 2024.

[10] Y. Hu et al., “Planning-oriented Autonomous Driving,” IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023.